Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: James J. Oliveira

Physical address of dairy:

730 W Monte Vista	Ceres	Stanislaus	95307
Number and Street	City	County	Zip Code

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1936

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

0022-0023-0003-0000

B. OPERATORS

Oliveira, Joseph

Operator name: Oliveira, Joseph

Telephone no.: (209) 595-6910 (Landline) (Cellular)

730 W Monte Vista AVE	Ceres	CA	95307
Mailing Address Number and Street	City	State	Zip Code

This operator is responsible for paying permit fees.

same as owner

Operator name: same as owner

Telephone no.: (209) 537-2310 (Landline) (Cellular)

730 W Monte Vista AVE	Ceres	CA	95307
Mailing Address Number and Street	City	State	Zip Code

C. OWNERS

Oliveira, Joseph

Legal owner name: Oliveira, Joseph

Telephone no.: (209) 595-6910
Landline | Cellular

730 W Monte Vista AVE | Ceres | CA | 95307
Mailing Address Number and Street | City | State | Zip Code

This owner is responsible for paying permit fees.

same as owner

Legal owner name: same as owner

Telephone no.: (209) 537-2310
Landline | Cellular

730 W Monte Vista AVE | Ceres | CA | 95307
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AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	25	25	35	0
Number under roof	0	0	0	0	0	0
Maximum number	0	0	25	25	35	0
Average number	0	0	0	0	0	0
Avg live weight (lbs)	0	0	900	700		

Predominant milk cow breed: Jersey-Holstein Cross

Average milk production: 1 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 1.00 *tons per reporting period*

Total nitrogen from manure: 1.00 *lbs per reporting period*

After ammonia losses (30% loss applied): 0.70 *lbs per reporting period*

Total phosphorus from manure: 1.00 *lbs per reporting period*

Total potassium from manure: 1.00 *lbs per reporting period*

Total salt from manure: 0.00 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 368,000 *gallons*

Total nitrogen generated: 932.34 *lbs*

Total phosphorus generated: 164.82 *lbs*

Total potassium generated: 1,989.98 *lbs*

Total salt generated: 10,809.78 *lbs*

	2,280,400	*gallons applied*
+	0	*gallons exported*
-	1,912,400	*gallons imported*
=	368,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
TID	Surface water
Well	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

Date	Material type / Description	Quantity	N (mg/L)	P (mg/L)	K (mg/L)	Salt (mg/L)	TDS (mg/L)
04/02/2023	Process wastewater *ww*	368,000.00 *gal*	477.80	59.70	854.00		4,040
05/22/2023	Process wastewater *ww*	432,000.00 *gal*	477.80	59.70	854.00		4,040
07/10/2023	Process wastewater *ww*	176,400.00 *gal*	178.40	38.00	400.00		2,700
07/19/2023	Process wastewater *ww*	288,000.00 *gal*	178.40	38.00	400.00		2,700
08/10/2023	Process wastewater *ww*	360,000.00 *gal*	178.40	38.00	400.00		2,700
08/21/2023	Process wastewater *ww*	288,000.00 *gal*	178.40	38.00	400.00		2,700

No commercial or other nutrient imports entered.

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	4,845.88	751.31	9,414.50	52,035.08
Total imports for all materials	4,845.88	751.31	9,414.50	52,035.08

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
Don Juan	8	8	7	none	0022-0023-0008-0000
Joe Luis	11	11	7	none	0022-0003-0022-0000
Lagoon	10	10	7	none	0022-0023-0003-0000
Milk Barn	20	20	7	none	0022-0023-0003-0000
The 50 #1	45	45	3	process wastewater	0022-0022-0015-0000 0022-0027-0015-0000
Zumstein	13	13	7	none	0022-0023-0007-0000
Totals for areas that were used for application	45	45	3		
Totals for areas that were not used for application	62	62	35		
Land application area totals	107	107	38		

B. CROPS AND HARVESTS

Don Juan

Field name: Don Juan

09/30/2021: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 8 Plant date: 09/30/2021

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/23/2023	8.81 *ton*	Dry-weight		8.8	29,900.00	5,600.00	43,000.00		9.20
05/21/2023	9.21 *ton*	Dry-weight		8.8	29,900.00	5,600.00	43,000.00		9.20
06/17/2023	10.16 *ton*	Dry-weight		8.8	29,900.00	5,600.00	43,000.00		9.20
07/28/2023	10.76 *ton*	Dry-weight		8.8	29,900.00	5,600.00	43,000.00		9.20
08/26/2023	12.95 *ton*	Dry-weight		8.8	29,900.00	5,600.00	43,000.00		9.20
09/15/2023	9.32 *ton*	Dry-weight		8.8	29,900.00	5,600.00	43,000.00		9.20
10/08/2023	8.99 *ton*	Dry-weight		8.8	29,900.00	5,600.00	43,000.00		9.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	480.00	43.20	336.00	8.00
Total actual harvest content	8.78	478.57	89.63	688.24	1,472.52

Joe Luis

Field name: Joe Luis

09/30/2021: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 11 Plant date: 09/30/2021

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/23/2023	9.32 *ton*	Dry-weight		8.8	29,900.00	5,600.00	43,000.00		9.20
05/21/2023	10.72 *ton*	Dry-weight		8.8	29,900.00	5,600.00	43,000.00		9.20
06/17/2023	13.61 *ton*	Dry-weight		8.8	29,900.00	5,600.00	43,000.00		9.20
07/28/2023	12.92 *ton*	Dry-weight		8.8	29,900.00	5,600.00	43,000.00		9.20
08/26/2023	14.01 *ton*	Dry-weight		8.8	29,900.00	5,600.00	43,000.00		9.20
09/15/2023	11.01 *ton*	Dry-weight		8.8	29,900.00	5,600.00	43,000.00		9.20
10/08/2023	8.46 *ton*	Dry-weight		8.8	29,900.00	5,600.00	43,000.00		9.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	480.00	43.20	336.00	8.00
Total actual harvest content	7.28	396.88	74.33	570.77	1,221.18

Lagoon

Field name: Lagoon

Lagoon

09/30/2021: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 10 Plant date: 09/30/2021

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/23/2023	12.31 *ton*	Dry-weight		11.9	36,000.00	5,600.00	44,000.00		9.88
05/21/2023	14.18 *ton*	Dry-weight		11.9	36,000.00	5,600.00	44,000.00		9.88
06/17/2023	15.26 *ton*	Dry-weight		11.9	36,000.00	5,600.00	44,000.00		9.88
07/28/2023	21.65 *ton*	Dry-weight		11.9	36,000.00	5,600.00	44,000.00		9.88
08/26/2023	13.82 *ton*	Dry-weight		11.9	36,000.00	5,600.00	44,000.00		9.88
09/15/2023	11.02 *ton*	Dry-weight		11.9	36,000.00	5,600.00	44,000.00		9.88
10/08/2023	90.30 *ton*	Dry-weight		11.9	36,000.00	5,600.00	44,000.00		9.88

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	1.40	84.00	7.56	58.80	1.40
Total actual harvest content	17.85	1,132.51	176.17	1,384.18	3,108.12

Milk Barn

Field name: Milk Barn

09/30/2021: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 20 Plant date: 09/30/2021

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/23/2023	17.98 *ton*	Dry-weight		11.9	36,000.00	5,600.00	44,000.00		9.88
05/21/2023	20.64 *ton*	Dry-weight		11.9	36,000.00	5,600.00	44,000.00		9.88
06/17/2023	21.23 *ton*	Dry-weight		11.9	36,000.00	5,600.00	44,000.00		9.88
07/28/2023	35.54 *ton*	Dry-weight		11.9	36,000.00	5,600.00	44,000.00		9.88
08/26/2023	24.60 *ton*	Dry-weight		11.9	36,000.00	5,600.00	44,000.00		9.88
09/15/2023	18.00 *ton*	Dry-weight		11.9	36,000.00	5,600.00	44,000.00		9.88
10/08/2023	17.60 *ton*	Dry-weight		11.9	36,000.00	5,600.00	44,000.00		9.88

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	1.10	66.00	5.94	46.20	1.10
Total actual harvest content	7.78	493.47	76.76	603.13	1,354.30

The 50 #1

Field name: The 50 #1

06/07/2022: Corn, silage

Crop: Corn, silage Acres planted: 45 Plant date: 06/07/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/26/2023	1,584.00 *ton*	Dry-weight		81.9	11,500.00	1,800.00	18,600.00		4.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	30.00
Total actual harvest content	35.20	146.54	22.94	237.01	586.15

10/28/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 33 Plant date: 10/28/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/02/2023	875.00 *ton*	Dry-weight		78.2	18,600.00	2,200.00	59,000.00		15.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	22.00	220.00	37.40	165.00	22.00
Total actual harvest content	26.52	215.03	25.43	682.08	1,757.21

10/28/2022: Oats, hay

Crop: Oats, hay Acres planted: 12 Plant date: 10/28/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/21/2023	88.00 *ton*	Dry-weight		78.2	18,600.00	2,200.00	59,000.00		15.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	4.00	160.00	26.00	132.00	4.00
Total actual harvest content	7.33	59.47	7.03	188.64	485.99

Zumstein

Field name: Zumstein

Zumstein

09/30/2021: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 13 Plant date: 09/30/2021

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/23/2023	13.96 *ton*	Dry-weight		11.6	32,100.00	4,300.00	31,000.00		9.36
05/21/2023	16.17 *ton*	Dry-weight		11.6	32,100.00	4,300.00	31,000.00		9.36
06/17/2023	15.23 *ton*	Dry-weight		11.6	32,100.00	4,300.00	31,000.00		9.36
07/10/2023	16.32 *ton*	Dry-weight		11.6	32,100.00	4,300.00	31,000.00		9.36
08/08/2023	17.12 *ton*	Dry-weight		11.6	32,100.00	4,300.00	31,000.00		9.36
09/06/2023	15.45 *ton*	Dry-weight		11.6	32,100.00	4,300.00	31,000.00		9.36
10/02/2023	13.96 *ton*	Dry-weight		11.6	32,100.00	4,300.00	31,000.00		9.36

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	480.00	43.20	336.00	8.00
Total actual harvest content	8.32	472.40	63.28	456.21	1,377.47

NUTRIENT BUDGET

A. LAND APPLICATIONS

Don Juan - 09/30/2021: Alfalfa, hay

Field name: Don Juan

Crop: Alfalfa, hay — Plant date: 09/30/2021

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID	Surface water	15.35	0.00	0.00	1,005.57	1,401,155.00 *gal*
Application event totals		15.35	0.00	0.00	1,005.57	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID	Surface water	7.67	0.00	0.00	502.78	700,577.00 *gal*
Application event totals		7.67	0.00	0.00	502.78	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID	Surface water	11.49	0.00	0.00	753.01	1,049,237.00 *gal*
Application event totals		11.49	0.00	0.00	753.01	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID	Surface water	7.67	0.00	0.00	502.78	700,577.00 *gal*
Application event totals		7.67	0.00	0.00	502.78	

Joe Luis - 09/30/2021: Alfalfa, hay

Field name: Joe Luis

Crop: Alfalfa, hay — Plant date: 09/30/2021

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Joe Luis - 09/30/2021: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID	Surface water	13.94	0.00	0.00	913.30	1,749,814.00 *gal*
Application event totals		13.94	0.00	0.00	913.30	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID	Surface water	11.16	0.00	0.00	731.32	1,401,155.00 *gal*
Application event totals		11.16	0.00	0.00	731.32	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID	Surface water	8.36	0.00	0.00	547.64	1,049,237.00 *gal*
Application event totals		8.36	0.00	0.00	547.64	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID	Surface water	11.86	0.00	0.00	777.24	1,489,134.00 *gal*
Application event totals		11.86	0.00	0.00	777.24	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID	Surface water	8.36	0.00	0.00	547.64	1,049,237.00 *gal*
Application event totals		8.36	0.00	0.00	547.64	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID	Surface water	5.58	0.00	0.00	365.66	700,577.00 *gal*
Application event totals		5.58	0.00	0.00	365.66	

Lagoon - 09/30/2021: Alfalfa, hay

Lagoon - 09/30/2021: Alfalfa, hay

Field name: Lagoon

Crop: Alfalfa, hay

Plant date: 09/30/2021

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID	Surface water	13.09	0.00	0.00	857.46	1,493,479.00 *gal*
Application event totals		13.09	0.00	0.00	857.46	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID	Surface water	12.21	0.00	0.00	800.09	1,393,551.00 *gal*
Application event totals		12.21	0.00	0.00	800.09	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID	Surface water	10.47	0.00	0.00	685.97	1,194,783.00 *gal*
Application event totals		10.47	0.00	0.00	685.97	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID	Surface water	8.94	0.00	0.00	585.57	1,019,910.00 *gal*
Application event totals		8.94	0.00	0.00	585.57	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID	Surface water	11.34	0.00	0.00	742.72	1,293,624.00 *gal*
Application event totals		11.34	0.00	0.00	742.72	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID	Surface water	3.71	0.00	0.00	243.21	423,605.00 *gal*
Application event totals		3.71	0.00	0.00	243.21	

Lagoon - 09/30/2021: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID	Surface water	9.59	0.00	0.00	628.60	1,094,856.00 *gal*
Application event totals		9.59	0.00	0.00	628.60	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID	Surface water	13.31	0.00	0.00	871.80	1,518,461.00 *gal*
Application event totals		13.31	0.00	0.00	871.80	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID	Surface water	2.62	0.00	0.00	171.49	298,695.00 *gal*
Application event totals		2.62	0.00	0.00	171.49	

Milk Barn - 09/30/2021: Alfalfa, hay

Field name: Milk Barn

Crop: Alfalfa, hay

Plant date: 09/30/2021

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/25/2023	Surface (irrigation)	Standing water	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID	Surface water	13.09	0.00	0.00	857.46	2,986,958.00 *gal*
Application event totals		13.09	0.00	0.00	857.46	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID	Surface water	12.21	0.00	0.00	800.09	2,787,103.00 *gal*
Application event totals		12.21	0.00	0.00	800.09	

Milk Barn - 09/30/2021: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID	Surface water	10.47	0.00	0.00	685.97	2,389,566.00 *gal*
Application event totals		10.47	0.00	0.00	685.97	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID	Surface water	8.94	0.00	0.00	585.57	2,039,820.00 *gal*
Application event totals		8.94	0.00	0.00	585.57	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID	Surface water	11.34	0.00	0.00	742.72	2,587,248.00 *gal*
Application event totals		11.34	0.00	0.00	742.72	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID	Surface water	3.71	0.00	0.00	243.21	847,210.00 *gal*
Application event totals		3.71	0.00	0.00	243.21	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID	Surface water	9.59	0.00	0.00	628.60	2,189,712.00 *gal*
Application event totals		9.59	0.00	0.00	628.60	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID	Surface water	13.31	0.00	0.00	871.80	3,036,922.00 *gal*
Application event totals		13.31	0.00	0.00	871.80	

Milk Barn - 09/30/2021: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID	Surface water	2.62	0.00	0.00	171.49	597,391.00 *gal*
Application event totals		2.62	0.00	0.00	171.49	

The 50 #1 - 06/07/2022: Corn, silage

Field name: The 50 #1

Crop: Corn, silage

Plant date: 06/07/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
ww	Process wastewater	38.28	4.78	68.42	323.65	432,000.00 *gal*
TID	Surface water	0.98	0.00	0.00	64.30	504,000.00 *gal*
Application event totals		39.26	4.78	68.42	387.96	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID	Surface water	17.31	0.00	0.00	1,134.13	8,889,188.00 *gal*
Application event totals		17.31	0.00	0.00	1,134.13	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
ww	Process wastewater	5.84	1.24	13.08	88.32	176,400.00 *gal*
TID	Surface water	15.80	0.00	0.00	1,035.19	8,113,665.00 *gal*
Application event totals		21.63	1.24	13.08	1,123.51	

The 50 #1 - 06/07/2022: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
ww	Process wastewater	9.53	2.03	21.36	144.20	288,000.00 *gal*
TID	Surface water	14.66	0.00	0.00	960.36	7,527,135.00 *gal*
Application event totals		24.18	2.03	21.36	1,104.56	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID	Surface water	15.94	0.00	0.00	1,044.33	8,185,352.00 *gal*
Application event totals		15.94	0.00	0.00	1,044.33	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
ww	Process wastewater	11.91	2.54	26.70	180.25	360,000.00 *gal*
TID	Surface water	21.64	0.00	0.00	1,418.08	11,114,743.00 *gal*
Application event totals		33.55	2.54	26.70	1,598.34	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
ww	Process wastewater	9.53	2.03	21.36	144.20	288,000.00 *gal*
TID	Surface water	15.47	0.00	0.00	1,013.57	7,944,223.00 *gal*
Application event totals		25.00	2.03	21.36	1,157.77	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID	Surface water	18.74	0.00	0.00	1,227.68	9,622,350.00 *gal*
Application event totals		18.74	0.00	0.00	1,227.68	

The 50 #1 - 10/28/2022: Oats, silage-soft dough

Field name: The 50 #1

Crop: Oats, silage-soft dough

Plant date: 10/28/2022

The 50 #1 - 10/28/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/03/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
ww	Process wastewater	20.27	3.59	43.26	235.00	264,000.00 *gal*
TID	Surface water	7.98	0.00	0.00	23.69	6,246,544.00 *gal*
Application event totals		28.25	3.59	43.26	258.69	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
ww	Process wastewater	31.90	3.99	57.01	269.71	264,000.00 *gal*
TID	Surface water	10.01	0.00	0.00	655.92	3,770,084.00 *gal*
Application event totals		41.91	3.99	57.01	925.63	

The 50 #1 - 10/28/2022: Oats, hay

Field name: The 50 #1

Crop: Oats, hay

Plant date: 10/28/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/03/2022	Shank	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
ww	Process wastewater	21.96	3.88	46.87	254.58	104,000.00 *gal*
TID	Surface water	8.32	0.00	0.00	24.73	2,370,450.00 *gal*
Application event totals		30.28	3.88	46.87	279.30	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
ww	Process wastewater	34.56	4.32	61.76	292.19	104,000.00 *gal*
TID	Surface water	10.83	0.00	0.00	709.35	1,482,617.00 *gal*
Application event totals		45.38	4.32	61.76	1,001.54	

Zumstein - 09/30/2021: Alfalfa, hay

Zumstein - 09/30/2021: Alfalfa, hay

Field name: Zumstein

Crop: Alfalfa, hay

Plant date: 09/30/2021

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID	Surface water	12.28	0.00	0.00	804.45	1,821,501.00 *gal*
Application event totals		12.28	0.00	0.00	804.45	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID	Surface water	16.36	0.00	0.00	1,072.12	2,427,582.00 *gal*
Application event totals		16.36	0.00	0.00	1,072.12	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID	Surface water	10.92	0.00	0.00	715.23	1,619,474.00 *gal*
Application event totals		10.92	0.00	0.00	715.23	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID	Surface water	6.83	0.00	0.00	447.56	1,013,393.00 *gal*
Application event totals		6.83	0.00	0.00	447.56	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID	Surface water	10.92	0.00	0.00	715.23	1,619,474.00 *gal*
Application event totals		10.92	0.00	0.00	715.23	

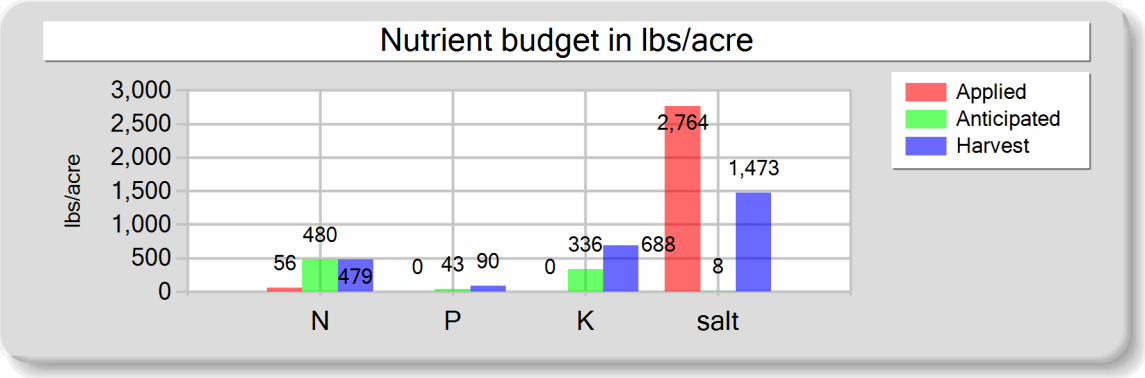
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID	Surface water	6.83	0.00	0.00	447.56	1,013,393.00 *gal*
Application event totals		6.83	0.00	0.00	447.56	

B. NUTRIENT BUDGET

Don Juan - 09/30/2021: Alfalfa, hay

Field name: Don Juan Crop: Alfalfa, hay Plant date: 09/30/2021



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	42.19	0.00	0.00	2,764.14
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	56.19	0.00	0.00	2,764.14
Anticipated crop nutrient removal	480.00	43.20	336.00	8.00
Actual crop nutrient removal	478.57	89.63	688.24	1,472.52
Nutrient balance	-422.38	-89.63	-688.24	1,291.62
Applied to removed ratio	0.12	0.00	0.00	1.88

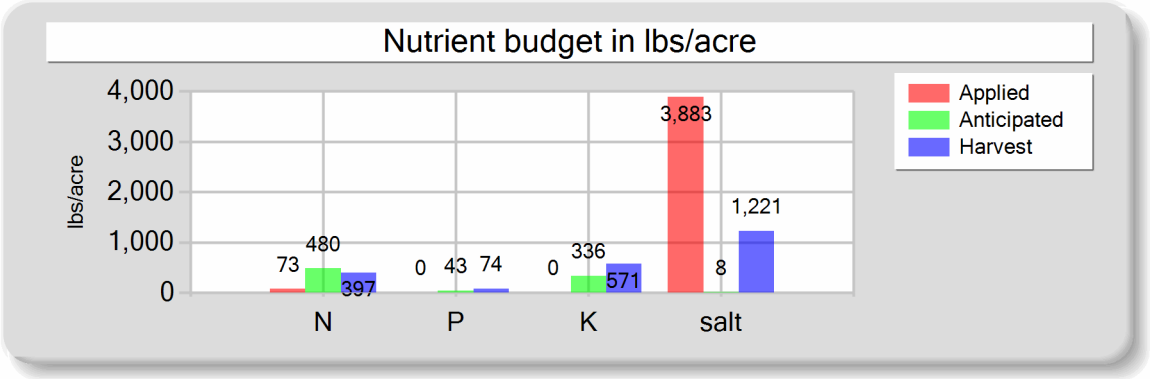
Fresh water applied	
3,851,546.00	*gallons*
141.84	*acre-inches*
17.73	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
7	*harvests*

Joe Luis - 09/30/2021: Alfalfa, hay

Field name: Joe Luis Crop: Alfalfa, hay Plant date: 09/30/2021



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	59.26	0.00	0.00	3,882.81
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	73.26	0.00	0.00	3,882.81
Anticipated crop nutrient removal	480.00	43.20	336.00	8.00
Actual crop nutrient removal	396.88	74.33	570.77	1,221.18
Nutrient balance	-323.63	-74.33	-570.77	2,661.62
Applied to removed ratio	0.18	0.00	0.00	3.18

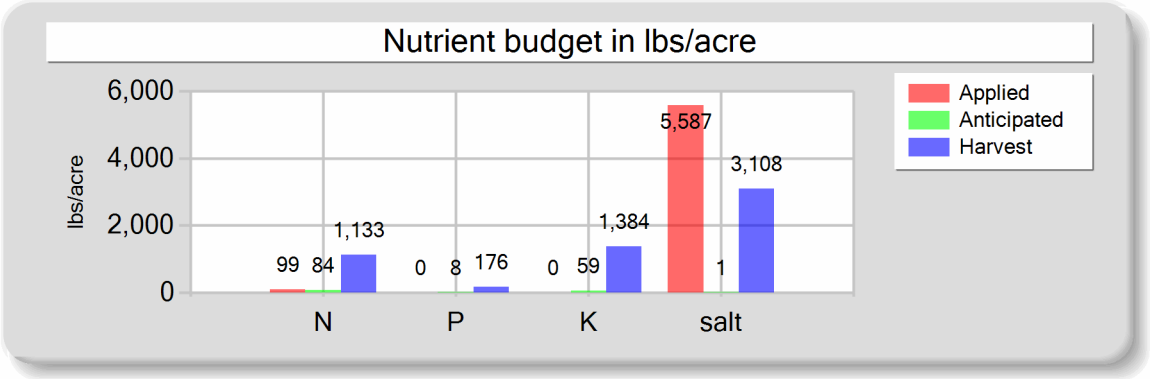
Fresh water applied
7,439,154.00 *gallons*
273.96 *acre-inches*
24.91 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
7 *harvests*

Lagoon - 09/30/2021: Alfalfa, hay

Field name: Lagoon Crop: Alfalfa, hay Plant date: 09/30/2021



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	85.27	0.00	0.00	5,586.90
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	99.27	0.00	0.00	5,586.90
Anticipated crop nutrient removal	84.00	7.56	58.80	1.40
Actual crop nutrient removal	1,132.51	176.17	1,384.18	3,108.12
Nutrient balance	-1,033.25	-176.17	-1,384.18	2,478.77
Applied to removed ratio	0.09	0.00	0.00	1.80

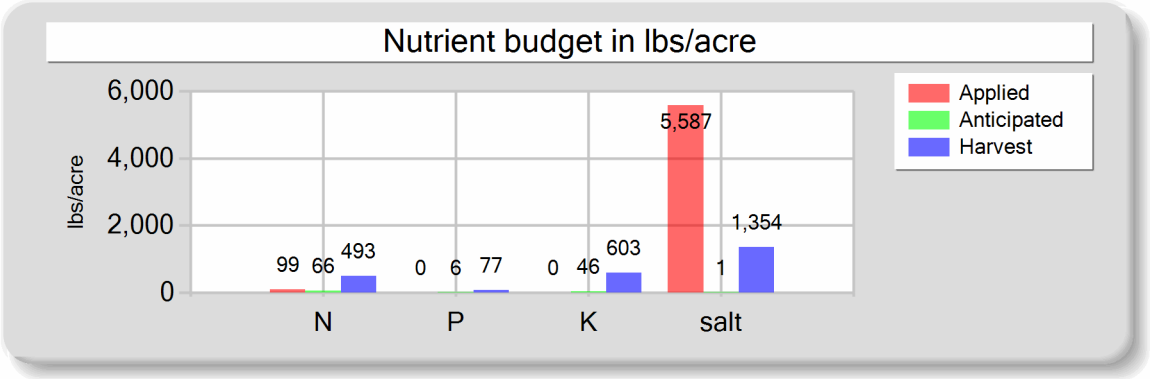
Fresh water applied	
9,730,964.00	*gallons*
358.36	*acre-inches*
35.84	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
7	*harvests*

Milk Barn - 09/30/2021: Alfalfa, hay

Field name: Milk Barn Crop: Alfalfa, hay Plant date: 09/30/2021



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	85.27	0.00	0.00	5,586.90
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	99.27	0.00	0.00	5,586.90
Anticipated crop nutrient removal	66.00	5.94	46.20	1.10
Actual crop nutrient removal	493.47	76.76	603.13	1,354.30
Nutrient balance	-394.20	-76.76	-603.13	4,232.60
Applied to removed ratio	0.20	0.00	0.00	4.13

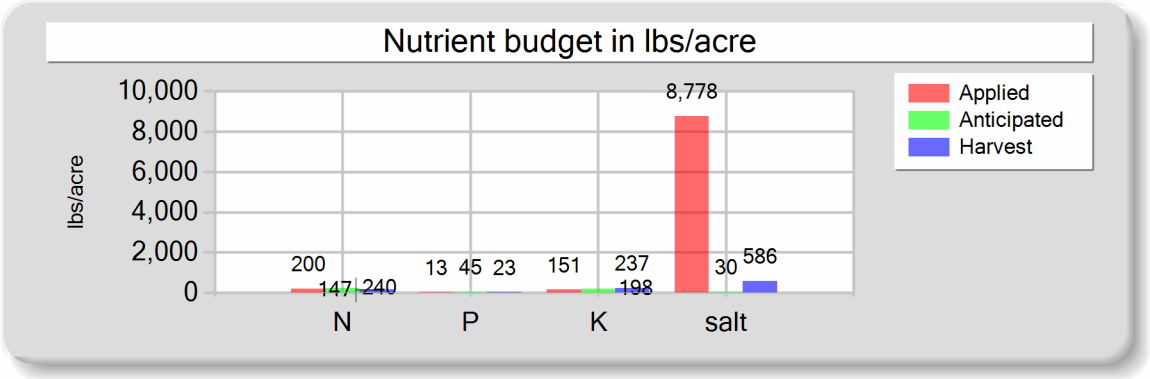
Fresh water applied	
19,461,930.00	*gallons*
716.72	*acre-inches*
35.84	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
7	*harvests*

The 50 #1 - 06/07/2022: Corn, silage

Field name: The 50 #1 Crop: Corn, silage Plant date: 06/07/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	75.08	12.62	150.93	880.63
Fresh water	120.53	0.00	0.00	7,897.64
Atmospheric deposition	4.67	0.00	0.00	0.00
Total nutrients applied	200.28	12.62	150.93	8,778.27
Anticipated crop nutrient removal	240.00	45.00	198.00	30.00
Actual crop nutrient removal	146.54	22.94	237.01	586.15
Nutrient balance	53.74	-10.31	-86.08	8,192.12
Applied to removed ratio	1.37	0.55	0.64	14.98

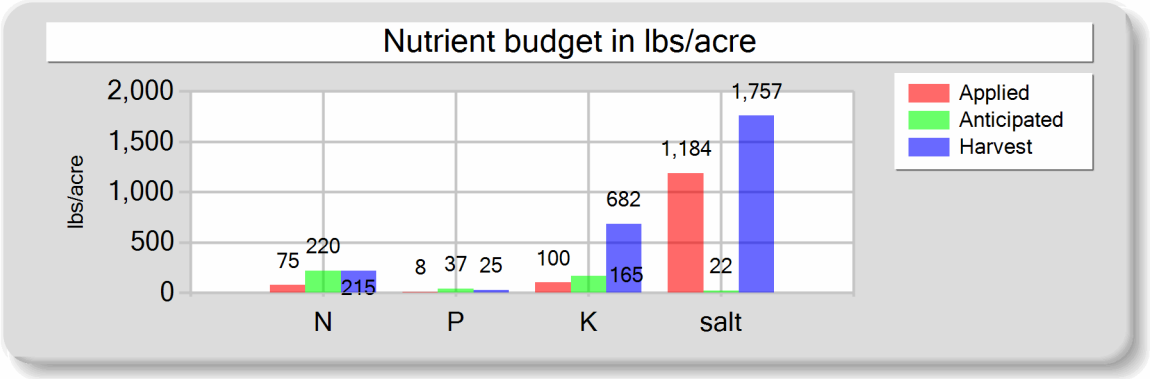
Fresh water applied	
61,900,656.00	*gallons*
2,279.59	*acre-inches*
50.66	*inches/acre*

Process wastewater applied	
1,544,400.00	*gallons*
56.88	*acre-inches*
1.26	*inches/acre*

Total harvests for the crop	
1	*harvests*

The 50 #1 - 10/28/2022: Oats, silage-soft dough

Field name: The 50 #1 Crop: Oats, silage-soft dough Plant date: 10/28/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	52.17	7.57	100.27	504.71
Fresh water	17.99	0.00	0.00	679.62
Atmospheric deposition	4.67	0.00	0.00	0.00
Total nutrients applied	74.82	7.57	100.27	1,184.32
Anticipated crop nutrient removal	220.00	37.40	165.00	22.00
Actual crop nutrient removal	215.03	25.43	682.08	1,757.21
Nutrient balance	-140.21	-17.86	-581.80	-572.89
Applied to removed ratio	0.35	0.30	0.15	0.67

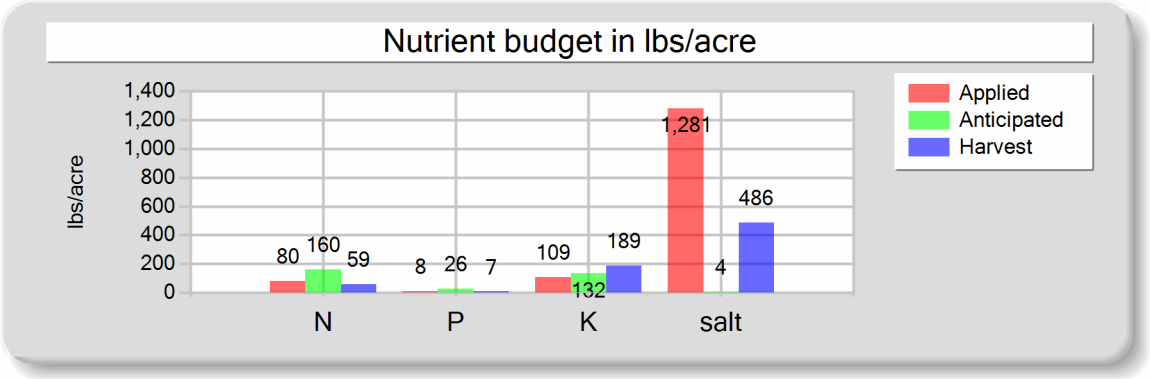
Fresh water applied	
10,016,628.00	*gallons*
368.88	*acre-inches*
11.18	*inches/acre*

Process wastewater applied	
528,000.00	*gallons*
19.44	*acre-inches*
0.59	*inches/acre*

Total harvests for the crop	
1	*harvests*

The 50 #1 - 10/28/2022: Oats, hay

Field name: The 50 #1 Crop: Oats, hay Plant date: 10/28/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	56.51	8.19	108.63	546.76
Fresh water	19.15	0.00	0.00	734.08
Atmospheric deposition	4.67	0.00	0.00	0.00
Total nutrients applied	80.33	8.19	108.63	1,280.84
Anticipated crop nutrient removal	160.00	26.00	132.00	4.00
Actual crop nutrient removal	59.47	7.03	188.64	485.99
Nutrient balance	20.86	1.16	-80.01	794.85
Applied to removed ratio	1.35	1.16	0.58	2.64

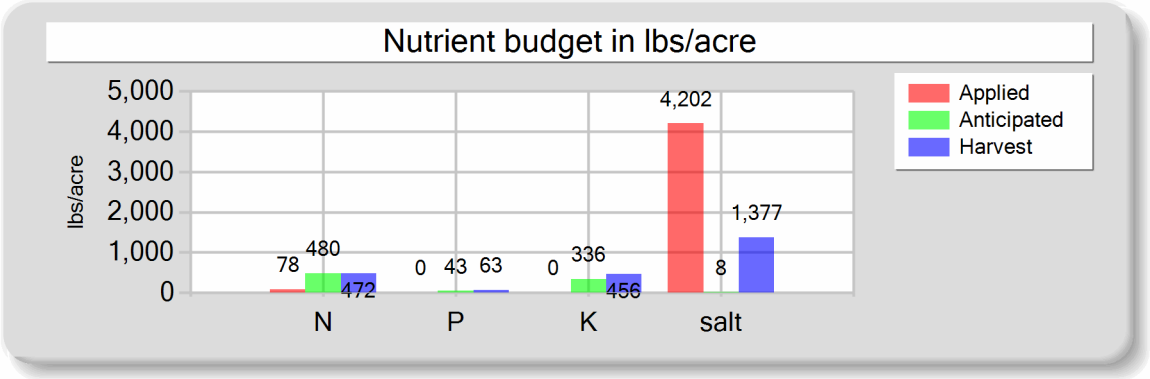
Fresh water applied	
3,853,067.00	*gallons*
141.90	*acre-inches*
11.82	*inches/acre*

Process wastewater applied	
208,000.00	*gallons*
7.66	*acre-inches*
0.64	*inches/acre*

Total harvests for the crop	
1	*harvests*

Zumstein - 09/30/2021: Alfalfa, hay

Field name: Zumstein Crop: Alfalfa, hay Plant date: 09/30/2021



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	64.13	0.00	0.00	4,202.15
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	78.13	0.00	0.00	4,202.15
Anticipated crop nutrient removal	480.00	43.20	336.00	8.00
Actual crop nutrient removal	472.40	63.28	456.21	1,377.47
Nutrient balance	-394.27	-63.28	-456.21	2,824.68
Applied to removed ratio	0.17	0.00	0.00	3.05

Fresh water applied
9,514,817.00 *gallons*
350.40 *acre-inches*
26.95 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
7 *harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

T3129011B

Sample and source description: T3129011B

Sample date: 05/09/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 24.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	20,200.00	5,500.00	11,800.00							
DL	0.01	0.01	0.01							

Manure

Sample and source description: Manure

Sample date: 09/26/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 6.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,000.00	4,570.00	27,600.00							
DL	0.01	0.01	0.01							

B. PROCESS WASTEWATER ANALYSES

ww

Sample and source description: ww

Sample date: 09/13/2022 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.40

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	302.00	243.00	0.00	1.60	53.70	648.00								7,310.00	3,520
DL	0.01	0.01	0.01	0.01	0.01	0.01								0.01	1

ww

Sample and source description: ww

Sample date: 02/16/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.50

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	702.00	205.00	0.00	1.70	96.40	1,090.00								10,400.00	5,150
DL	1.00	0.50	0.01	0.10	0.10	0.50								0.01	10

ww

Sample and source description: ww

Sample date: 06/20/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.40

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	477.00	375.00	0.00	0.80	59.70	854.00								8,850.00	4,040
DL	1.00	0.50	0.01	0.10	0.10	0.50								0.01	10

ww

Sample and source description: ww

Sample date: 09/14/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.40

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	178.00	153.00	0.00	0.40	38.00	400.00								5.22	2,700
DL	1.00	0.50	0.01	0.10	0.10	0.50								0.01	10

ww

Sample and source description: ww

Sample date: 11/09/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.50

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	366.00	280.00	0.00	0.70	53.60	639.00								6,530.00	3,420
DL	1.00	0.50	0.01	0.10	0.10	0.50								0.01	10

C. FRESH WATER ANALYSES

TID

canal

Sample description: canal

Sample date: 08/23/2022 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	5.05										613.00	15
DL	0.01										0.01	1

canal

Sample description: canal

Sample date: 06/20/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	10.50										100.00	688
DL	0.01										0.01	1

Well

well

Sample description: well

Sample date: 12/19/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	43.80										1,160.00	
DL	0.01										0.01	

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

Joe Luis - 09/30/2021: Alfalfa, hay

hay

Sample and source description: hay

Sample date: 05/09/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 8.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	29,900.00	5,600.00	43,000.00		9.20
DL	0.01	0.01	0.01		1.00

Milk Barn - 09/30/2021: Alfalfa, hay

Lagoon dry hay

Sample and source description: Lagoon dry hay

Sample date: 05/09/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 11.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	36,000.00	5,600.00	44,000.00		9.88
DL	0.01	0.01	0.01		1.00

Milk Barn

Sample and source description: Milk Barn

Sample date: 05/09/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 10.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	34,200.00	4,300.00	4.70		10.30
DL	0.01	0.01	0.01		1.00

The 50 #1 - 06/07/2022: Corn, silage

The 50 #1 - 06/07/2022: Corn, silage

T3269016B

Sample and source description: T3269016B

Sample date: 09/26/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 81.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,500.00	1,800.00	18,600.00		4.60
DL	0.01	0.01	0.01		1.00

The 50 #1 - 10/28/2022: Oats, silage-soft dough

winter forage

Sample and source description: winter forage

Sample date: 05/09/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 78.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	18,600.00	2,200.00	59,000.00		15.20
DL	0.01	0.01	0.01		1.00

Zumstein - 09/30/2021: Alfalfa, hay

hay

Sample and source description: hay

Sample date: 05/09/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 11.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	32,100.00	4,300.00	31,000.00		9.36
DL	0.01	0.01	0.01		1.00

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

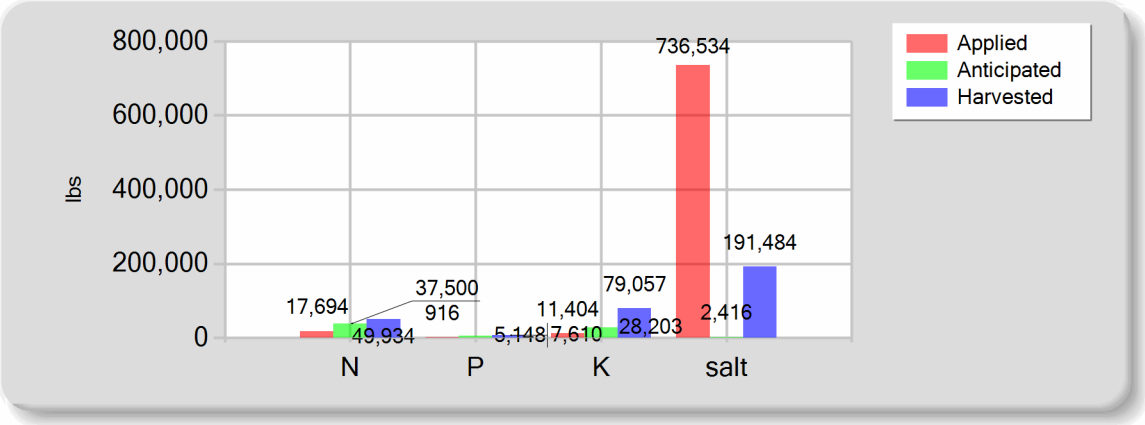
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

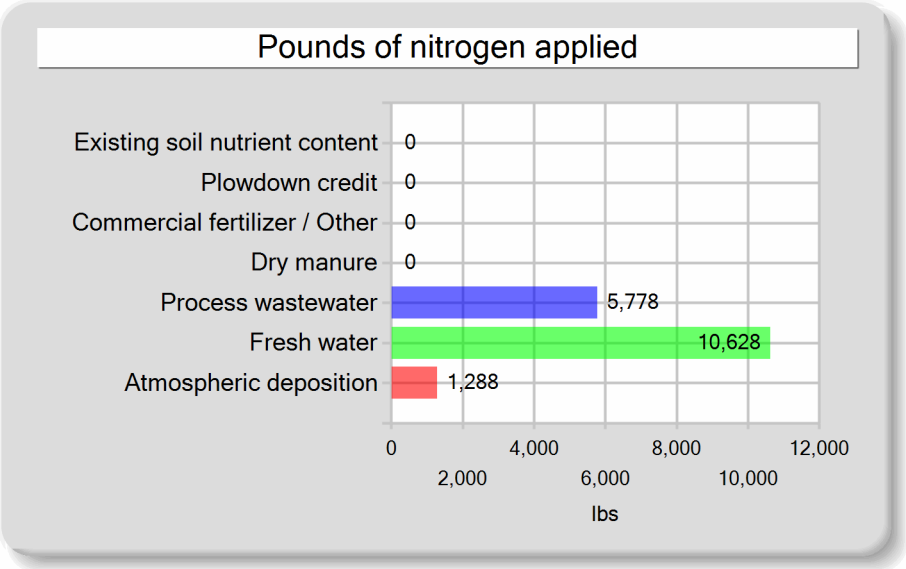
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

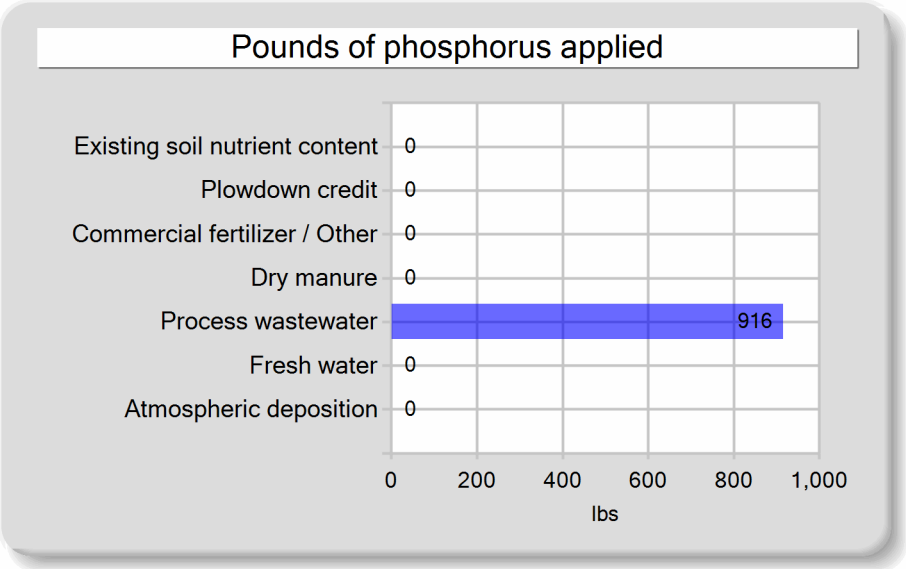
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	5,778.22	916.13	11,404.48	62,844.86
Fresh water	10,628.27	0.00	0.00	673,689.11
Atmospheric deposition	1,288.00	0.00	0.00	0.00
Total nutrients applied	17,694.49	916.13	11,404.48	736,533.97
Anticipated crop nutrient removal	37,500.00	5,148.00	28,203.00	2,416.00
Actual crop nutrient removal	49,933.77	7,610.14	79,057.21	191,484.19
Nutrient balance	-32,239.28	-6,694.01	-67,652.74	545,049.78
Applied to removed ratio	0.35	0.12	0.14	3.85

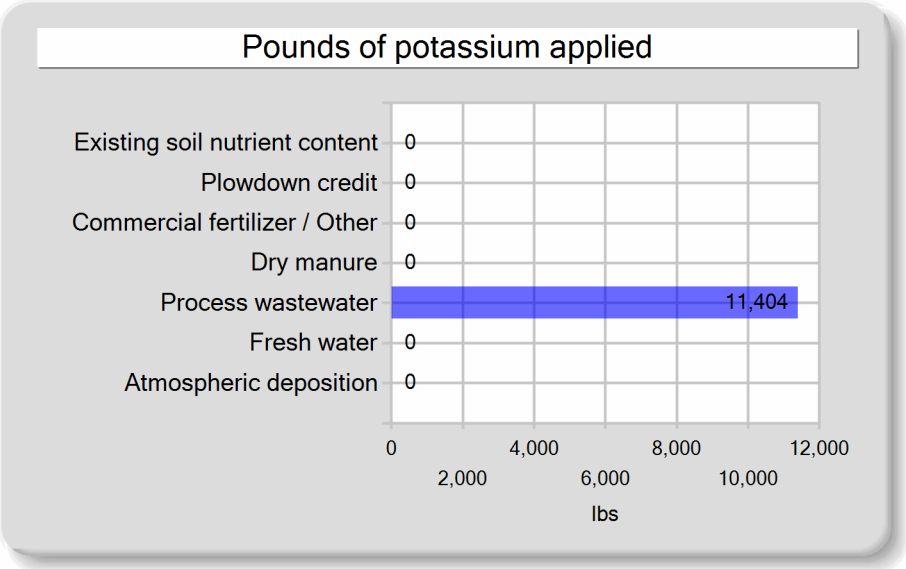
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

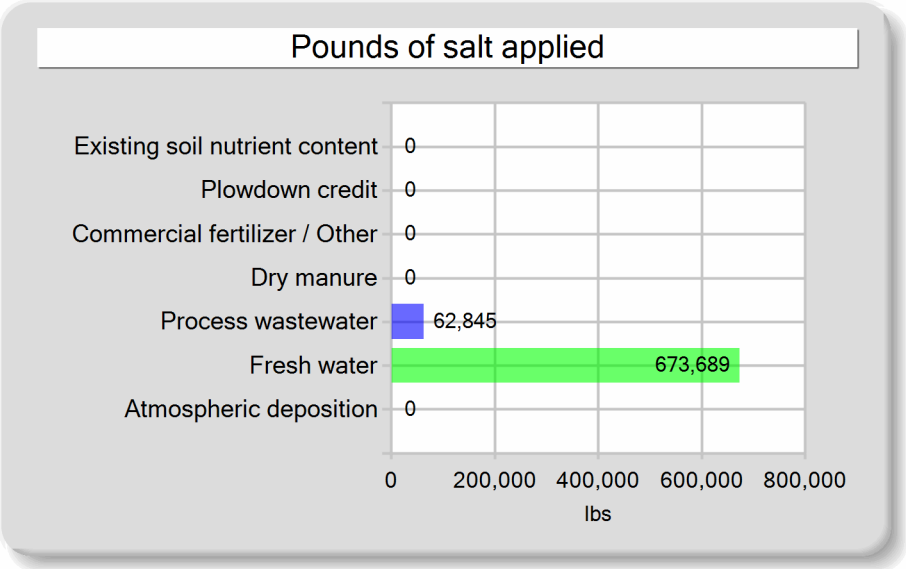


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	No
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Joseph Oliveira	same as owner
PRINT OR TYPE NAME	PRINT OR TYPE NAME
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Joseph Oliveira	Joseph Oliveira
SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Joseph Oliveira	same as owner
PRINT OR TYPE NAME	PRINT OR TYPE NAME
6/18/24	6/18/24
DATE	DATE

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

www.denelelabs.com

Compliance Analysis Report

Customer
OLIVEIRA HOLSTEINS
730 W Monte Vista Ave
Ceres, CA 95307

Grower

Consultant:
Afton Bettencourt
3140 W. Monte Vista Ave.
Modesto, CA 95358

PURCHASE ORDER:
RECEIVED DATE: 12/19/23 8:50 AM
SUBMITTED BY: Joey
ANALYZED DATE: 12/19/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Ranch Water	T3353001A	Electrical Conductivity (EC)	EPA 120.1 and	1.16	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	43.8	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	740	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	94.2	99.4 (T3352006A)	98.2 (T3352006A)	1.21
Electrical Conductivity (EC)	98.9			0.750
Nitrate Nitrogen (NO3-N)	105	106 (T3353008A)	100 (T3353008A)	5.20

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

DENELE ANALYTICAL, INC.

865 S. KILROY
TURLOCK, CA 95380

PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Client: Oliveira Holsteins
Mailing address:
(not facility address)
Phone#: (209) 535-2763

Authorized Copy Released To:

Sampled By:
Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
- H1 EC*, NO3-N, NH4-N*
- H6 EC, TN, TDS
- H7 Combination of H1 & H6
- H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
- H3 EC*, NH4-N, TKN, TP, K, TDS
- H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
- H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
- H8 Combination of H3 &H5
- H9 Combination of H4 &H5

TILE DRAIN
- H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*

___ (Check here if performed)

DISCHARGE WATER
- E1 EC*, Temp*, pH*, NO3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
- E2 EC,* Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
- E3 EC*, Temp*, pH*, NO3-N, NH4-N*, TP, Total & Fecal Coliforms
- E4 EC*, Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
- M1 % Moisture
- M3 TN TP, K, % Moisture
- M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
- M7 Combination of M3 & M4

PLANT TISSUE
- F18 % Moisture, Ash, TN, P, K, DM and as received
- F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
- F19+ % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
- K3 Soluble Phosphorus
- K4 1ft NO3, Organic Matter
- K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
- K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
- K7 Combination of K3 & K4
- K8 Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	Analysis: H1	H3	M3	F18	OTHER
1	12-18-23	4:10	Ranch Water		15.6°C		H-1				
2											
3											
4											
5											
6											
7											
8											
9											
10											

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	Logged in By:	LABORATORY NUMBER (LABORATORY USE ONLY)
1st						
2nd		DENELE ANALYTICAL, INC.	12-19-23 8:52			T3353001

Required to be submitted to any agency? (YES / NO)